

Win32 DIVX visual tool analysisor Implementation Guide

MMSP™ 2

User Manual

MDMP2520FMPEG4DUM

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MP2520F

*How to implement Divx visual tool analysis library in Win32
environment*

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1. Divx visual tool analysis library

Divx visual tool parser is a module which can parse Mpeg4 video streams can be played at “MMSP2 ARM940T firmware”. Divx visual tool parser parse “.avi” stream which be encoded Standard mpeg4 ASP(divx , xvid) has impossible visual tool denote MC warping point 2 or Qpel & 4MV or etc. Divx visual tool parser consist of visual_tool_parser.lib and mpeg4_error_header.h

1.1. Compiling environment

Application can link this library to parse mpeg4 stream. To use this library, you link the library on your application.

- mpeg4_error_header.h : define Function and declare error
- visual_tool_parser.lib

1.2. Library Interface

1.2.1. input parameter

visual_tool_parser.lib get avi file name and fream number to parse. and visual_tool_parser also get information structures (avi information struct, print struct) for user. If frame number is more than avi file’s frame number , visual_tool_parser library parse avi file using default frame number. When library parse end, avi information structure and print structure pointers has included mpeg4 video stream. User can know about parsed avi file information by these structure.

1.2.2. output parameter

visual_tool_parser.lib return error code.

MACRO	Description
DEBUG_PRINT_BY_CONSOLE	Internal header. ignored.
FILE_NAME_ERROR	File name pointer is 0
DECODER_INIT_ERROR	Decoder initialization failed.
DECODER_RUN_ERROR	Error occurred when decode process.

DECODER_RELEASE_ERROR	Memory Release error
GMC_WARPING_OVER_ONE	GMC warping point is 2 or more.
MV4_QPEL	Mpeg4 stream have 4MV & Qpel both.

2. MMSP2 mpeg4 decoder restrict

MMSP2 mpeg4 Decoder has restrict when mpeg4 video stream decoding. If mpeg4 video stream has both 4MV & Qpel, and stream's GMC(global motion estimation) warping point is 2 or more.